

Virtualization Technology

Definition: Virtualization ek aisi technology hai jisme ek **physical machine** ke upar multiple **virtual machines (VMs)** chal sakti hain. Matlab ek hi hardware ke upar alag-alag operating systems aur applications ek saath run ho sakte hain.

Kaise kaam karti hai:

- Ek **Hypervisor** use hota hai jo hardware aur virtual machines ke beech ek layer ka kaam karta hai.
- Hypervisor hardware resources (CPU, RAM, Storage, Network) ko alag-alag VMs me divide karke deta hai.
- Har VM apna alag **Operating System** aur **Applications** run kar sakta hai.

Types of Virtualization:

1. Server Virtualization
2. Desktop Virtualization
3. Network Virtualization
4. Storage Virtualization

Types of Hypervisors:

1. Type 1 (Bare Metal) – VMware ESXi, Microsoft Hyper-V, KVM
2. Type 2 (Hosted) – VirtualBox, VMware Workstation

Benefits of Virtualization:

- Hardware ka better utilization
- Cost saving
- Disaster Recovery easy hota hai
- Fast provisioning aur scalability
- Testing & Development me flexibility

Real Life Example:

Agar ek server me 64 GB RAM aur 16 CPU cores hain, toh usse 4 VMs bana sakte hain (16 GB RAM + 4 CPU cores har VM ko). Har VM apna alag OS aur Application run karega.

Virtualization Diagram:

Virtualization Technology Diagram

